

MoE Expert Routing Analysis — Run 2

Qwen3-VL-30B-A3B-Instruct

LEGOLite (400 questions)

HEIGHT

32.0%

↑ 0.0% vs Run 1

POSITION

25.0%

↑ 3.0% vs Run 1

ROTATION

24.0%

↑ 1.0% vs Run 1

ORDERING

20.0%

↑ 0.0% vs Run 1

OVERALL ACCURACY

25.2%

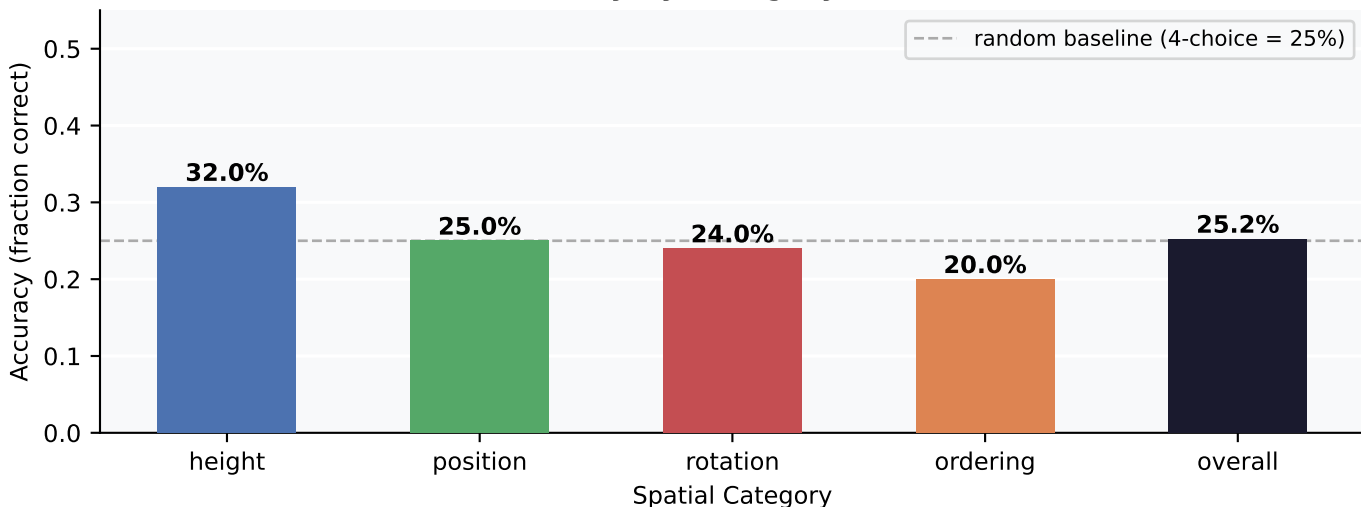
Key Findings

- 48 MoE layers × 128 experts × Top-8 routing analysed
- 400 LEGOLite questions (100 per spatial category)
- Router activations captured for visual tokens only
- Run 1 overall: 24.2% | Run 2 overall: 25.2%
- Δ overall: +1.0%
- See page 9 for full Run 1 vs Run 2 comparison

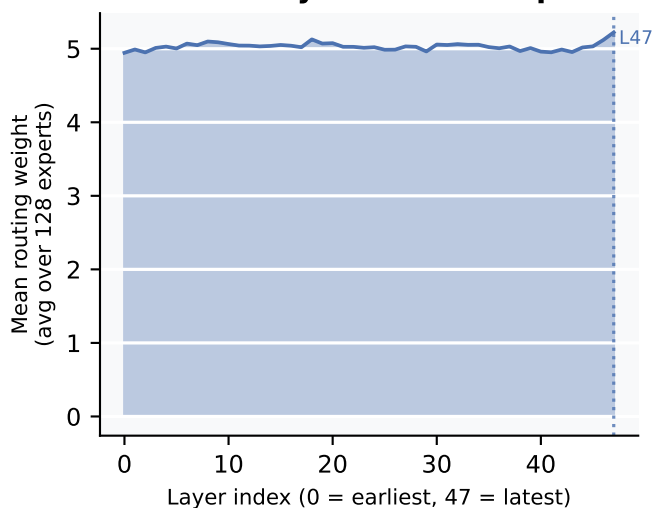
Accuracy & Per-Category Expert Activation

How often the model answers correctly, and which layers are most active per category

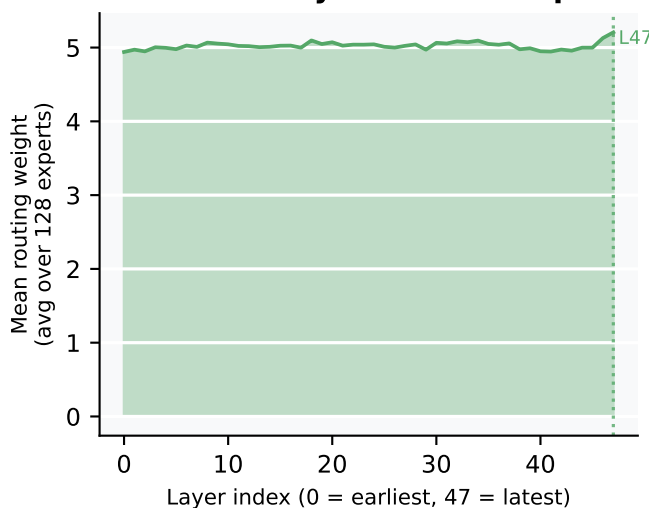
Accuracy by Category — Run 2



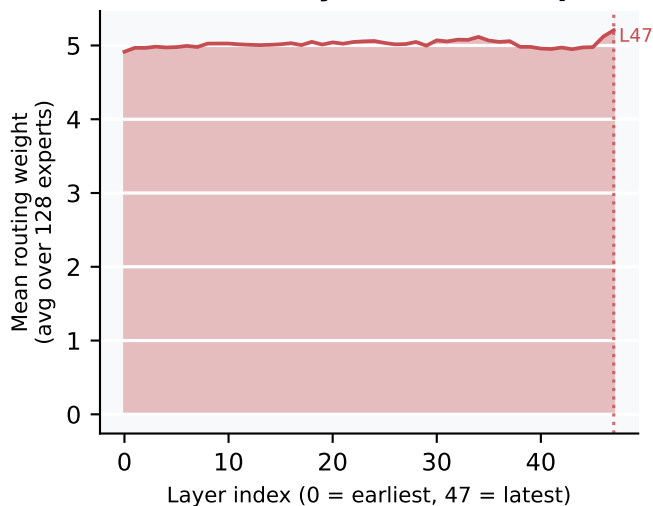
HEIGHT — layer activation profile



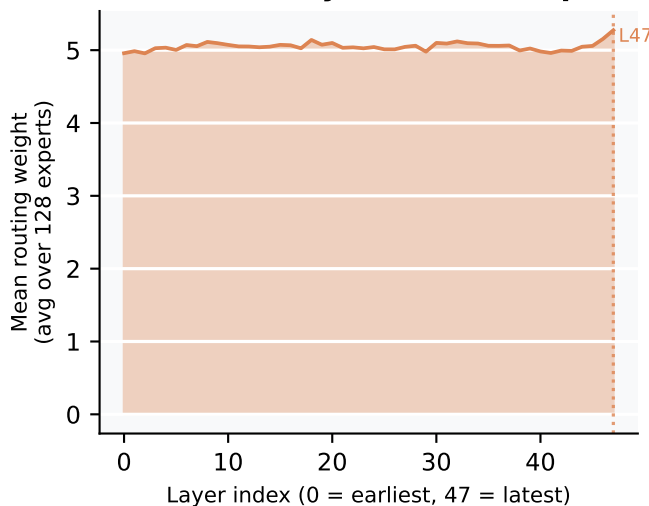
POSITION — layer activation profile



ROTATION — layer activation profile



ORDERING — layer activation profile



Top-30 Spatial Reasoning Experts — Global Leaderboard

Ranked by success_delta = mean_activation_correct – mean_activation_incorrect

Rank	Expert	Act (correct)	Act (incorrect)	Success Δ
1	L10_E34	5.50613	5.40634	+0.09979
2	L27_E50	5.06313	4.97635	+0.08678
3	L20_E118	5.45542	5.36917	+0.08625
4	L16_E3	5.66982	5.58713	+0.08269
5	L20_E115	5.69673	5.61656	+0.08017
6	L34_E1	5.07976	5.00565	+0.07412
7	L28_E71	5.56972	5.49822	+0.07150
8	L18_E120	5.10504	5.03360	+0.07144
9	L15_E50	5.12793	5.05660	+0.07134
10	L34_E22	5.49644	5.42535	+0.07109
11	L17_E56	5.61045	5.54043	+0.07002
12	L8_E115	5.55742	5.49097	+0.06645
13	L14_E83	5.54647	5.48016	+0.06630
14	L6_E100	5.36520	5.29972	+0.06548
15	L9_E56	5.22590	5.16065	+0.06526
16	L22_E34	5.36380	5.29878	+0.06502
17	L7_E20	6.36845	6.30344	+0.06501
18	L34_E34	5.41695	5.35322	+0.06373
19	L9_E103	5.60738	5.54378	+0.06361
20	L21_E56	5.39668	5.33334	+0.06333
21	L28_E53	5.18946	5.12618	+0.06328
22	L20_E95	6.19670	6.13377	+0.06294
23	L30_E120	5.30806	5.24540	+0.06265
24	L32_E115	5.50603	5.44341	+0.06262
25	L21_E55	5.25262	5.19048	+0.06214
26	L16_E71	5.46513	5.40366	+0.06147
27	L15_E91	5.90421	5.84348	+0.06074
28	L21_E26	4.91532	4.85463	+0.06069
29	L28_E3	5.28860	5.22815	+0.06045
30	L11_E76	4.97226	4.91227	+0.06000

Category Specialist Experts

Top-10 experts per spatial category, ranked by category-specific success Δ

HEIGHT specialists

Expert	Act (correct)	Δ height
L10_E34	5.50613	+0.14618
L16_E3	5.66982	+0.12386
L9_E103	5.60738	+0.12099
L20_E118	5.45542	+0.11329
L27_E50	5.06313	+0.11191
L8_E115	5.55742	+0.10306
L8_E70	5.54202	+0.10263
L15_E50	5.12793	+0.10125
L20_E115	5.69673	+0.09862
L11_E11	5.09893	+0.09526

POSITION specialists

Expert	Act (correct)	Δ position
L10_E36	4.94234	+0.11173
L23_E123	5.44011	+0.10577
L10_E6	5.80832	+0.09659
L4_E18	4.66350	+0.08777
L8_E114	5.23418	+0.08760
L34_E51	5.45891	+0.08614
L35_E123	5.59087	+0.07995
L35_E51	4.72863	+0.07949
L6_E100	5.36520	+0.07910
L34_E1	5.07976	+0.07883

ROTATION specialists

Expert	Act (correct)	Δ rotation
L6_E54	5.40664	+0.10935
L18_E54	5.15075	+0.10841
L6_E72	5.09926	+0.09502
L29_E112	6.11975	+0.08706
L17_E112	6.01217	+0.08627
L30_E127	5.83567	+0.08507
L6_E109	5.20497	+0.08089
L31_E32	5.09487	+0.07968
L18_E72	4.68295	+0.07916
L3_E9	5.38091	+0.07870

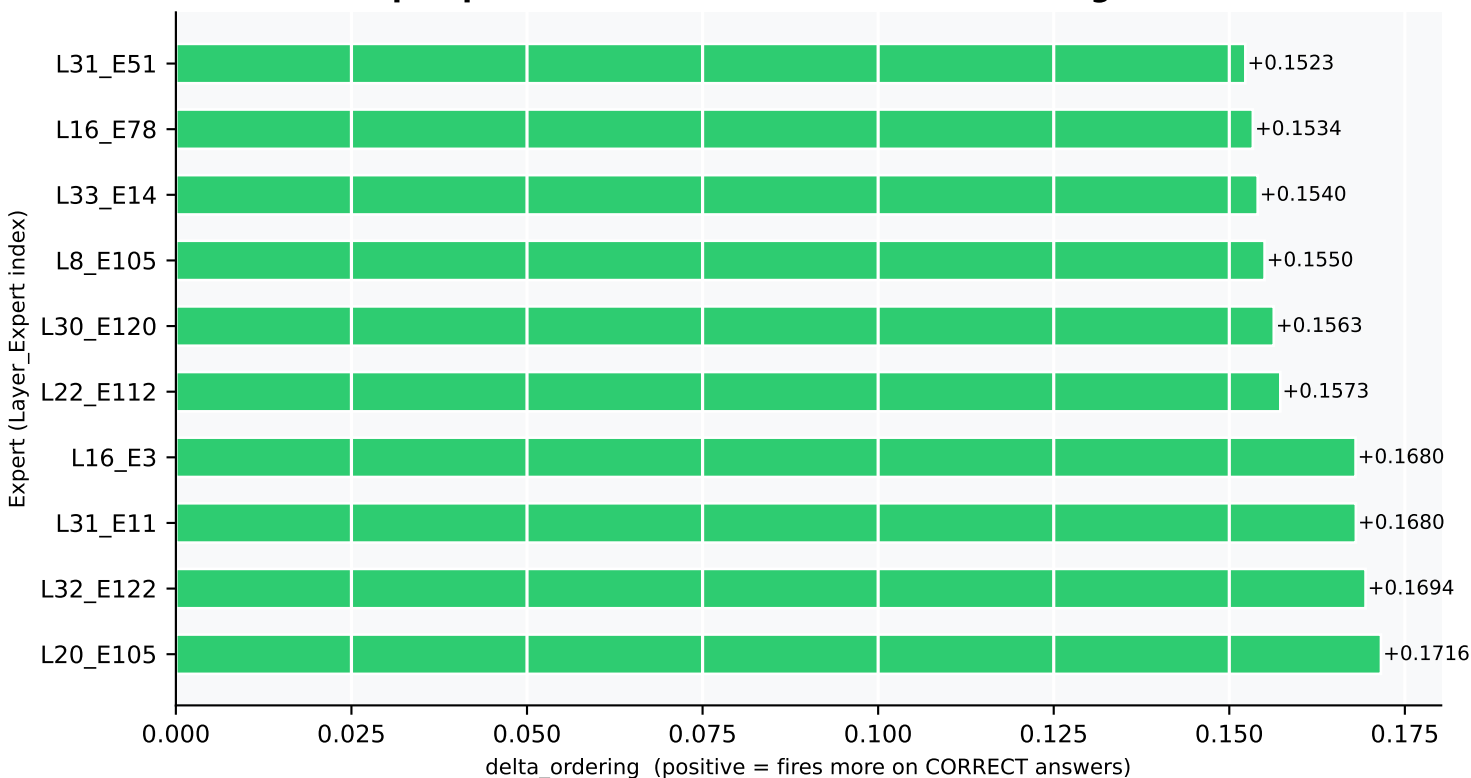
ORDERING specialists

Expert	Act (correct)	Δ ordering
L20_E105	5.91013	+0.17160
L32_E122	6.07552	+0.16941
L31_E11	5.26476	+0.16802
L16_E3	5.66982	+0.16799
L22_E112	5.37899	+0.15727
L30_E120	5.30806	+0.15634
L8_E105	5.95570	+0.15503
L33_E14	4.97648	+0.15405
L16_E78	5.35421	+0.15336
L31_E51	6.04491	+0.15229

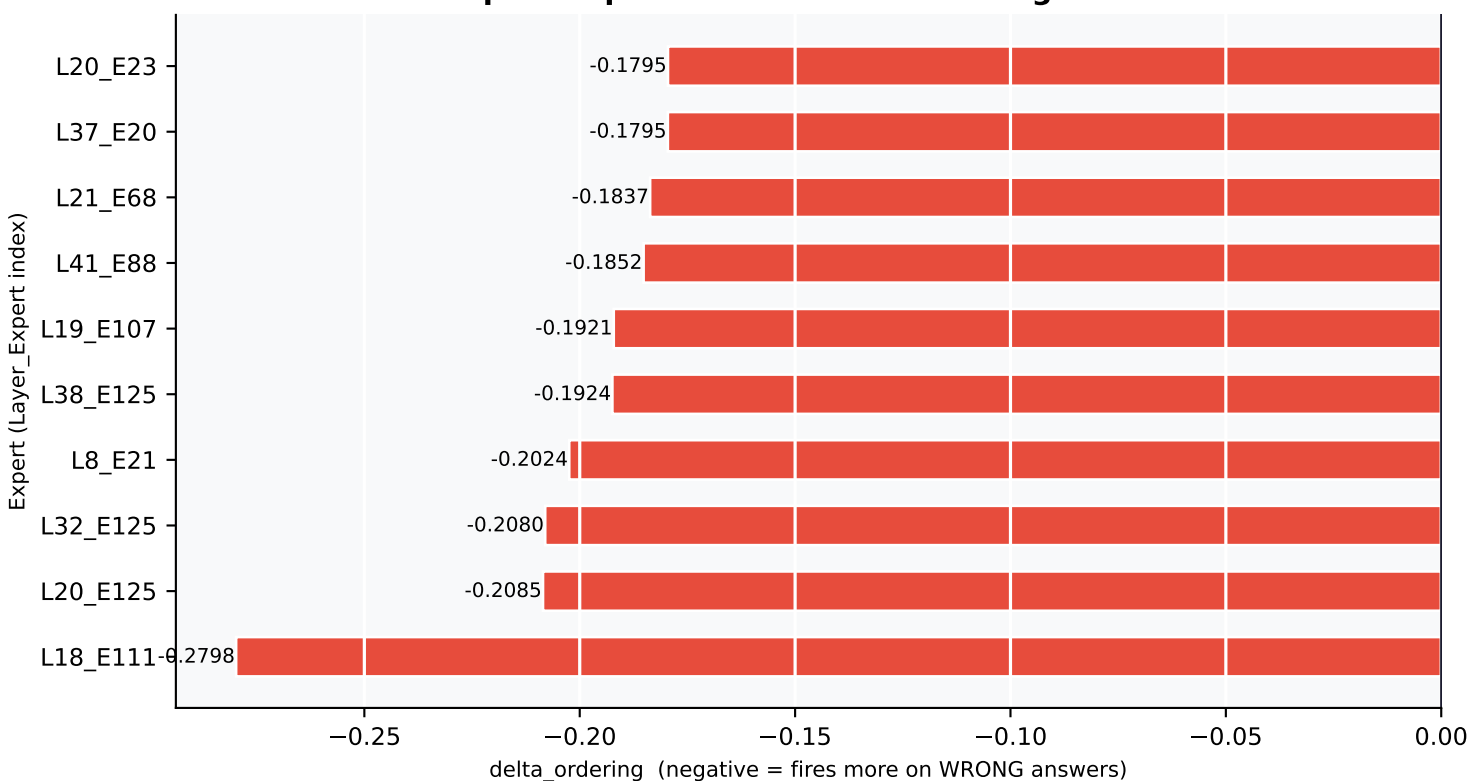
Ordering Task — Expert Failure Diagnosis

Experts that activate heavily on ordering tasks but correlate with WRONG answers

Top Experts Correlated With Correct Ordering Answers



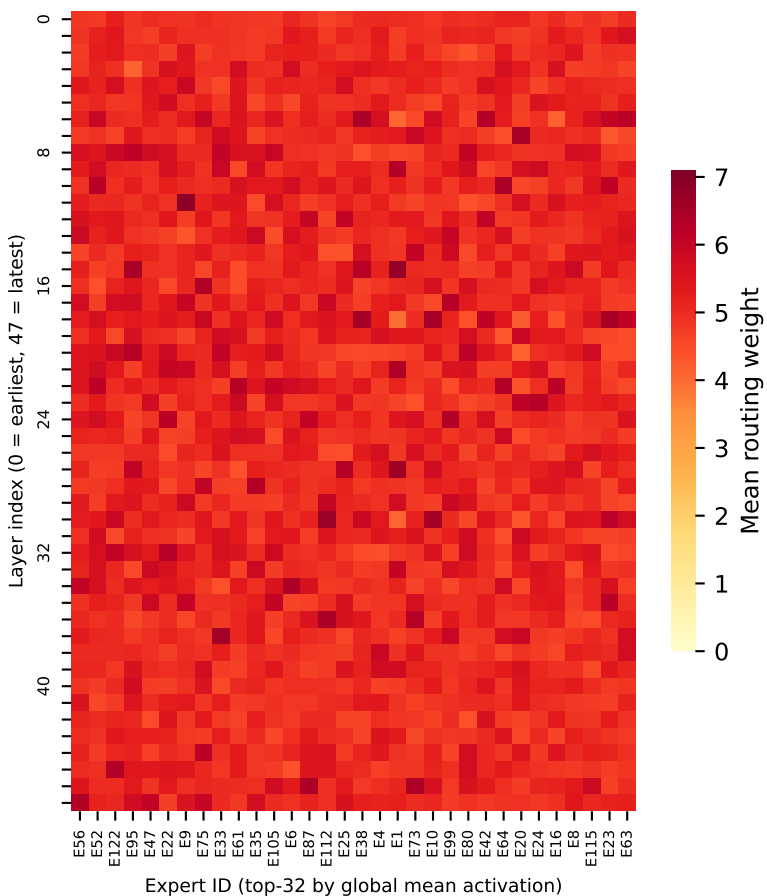
Suspect Experts — Active on Ordering Failures



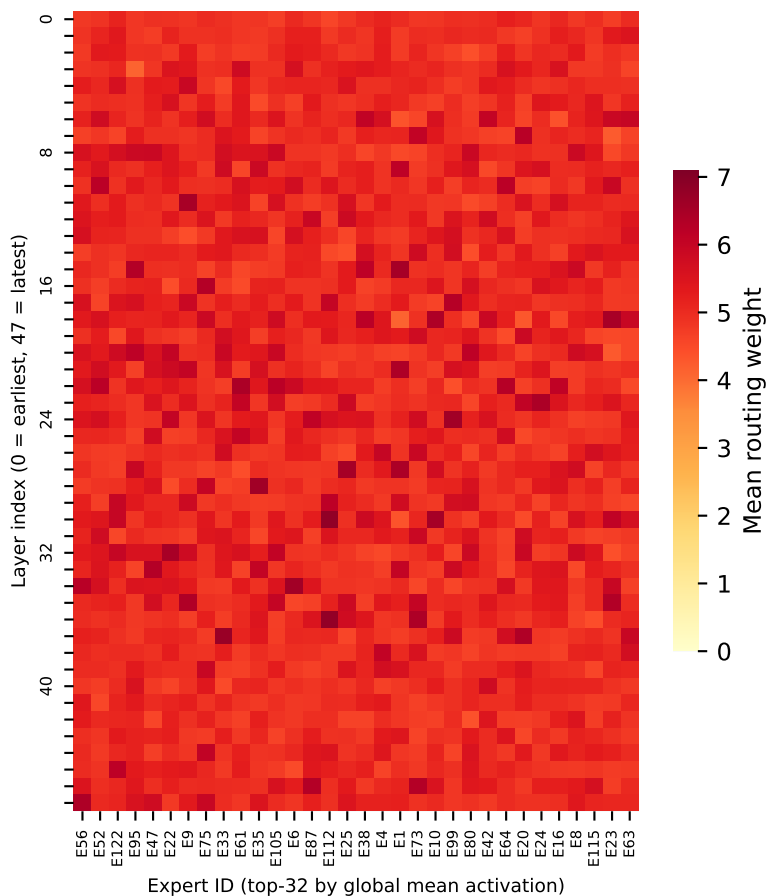
Visual Token Expert Activation Heatmaps — Run 2

(Layer × Top-32 most active experts; colour = mean routing weight)

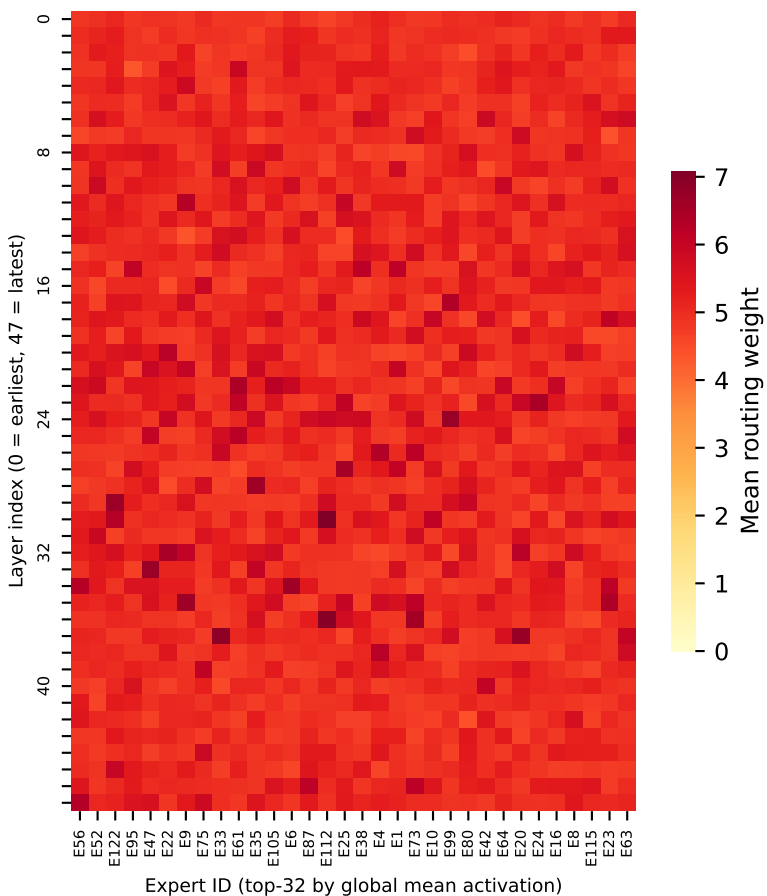
HEIGHT (acc=32.0%)



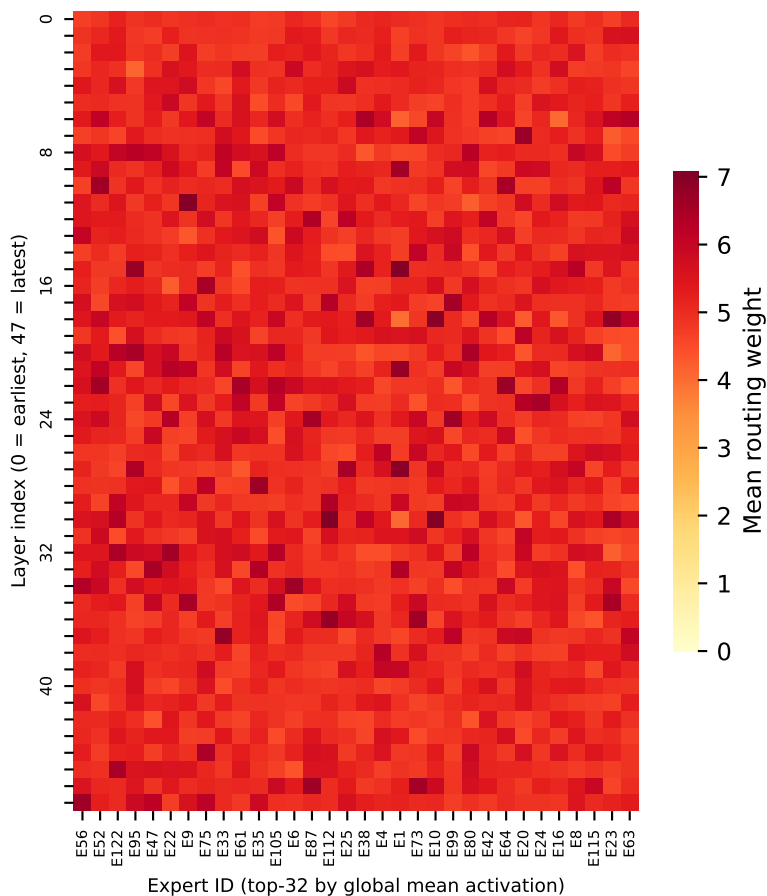
POSITION (acc=25.0%)



ROTATION (acc=24.0%)



ORDERING (acc=20.0%)

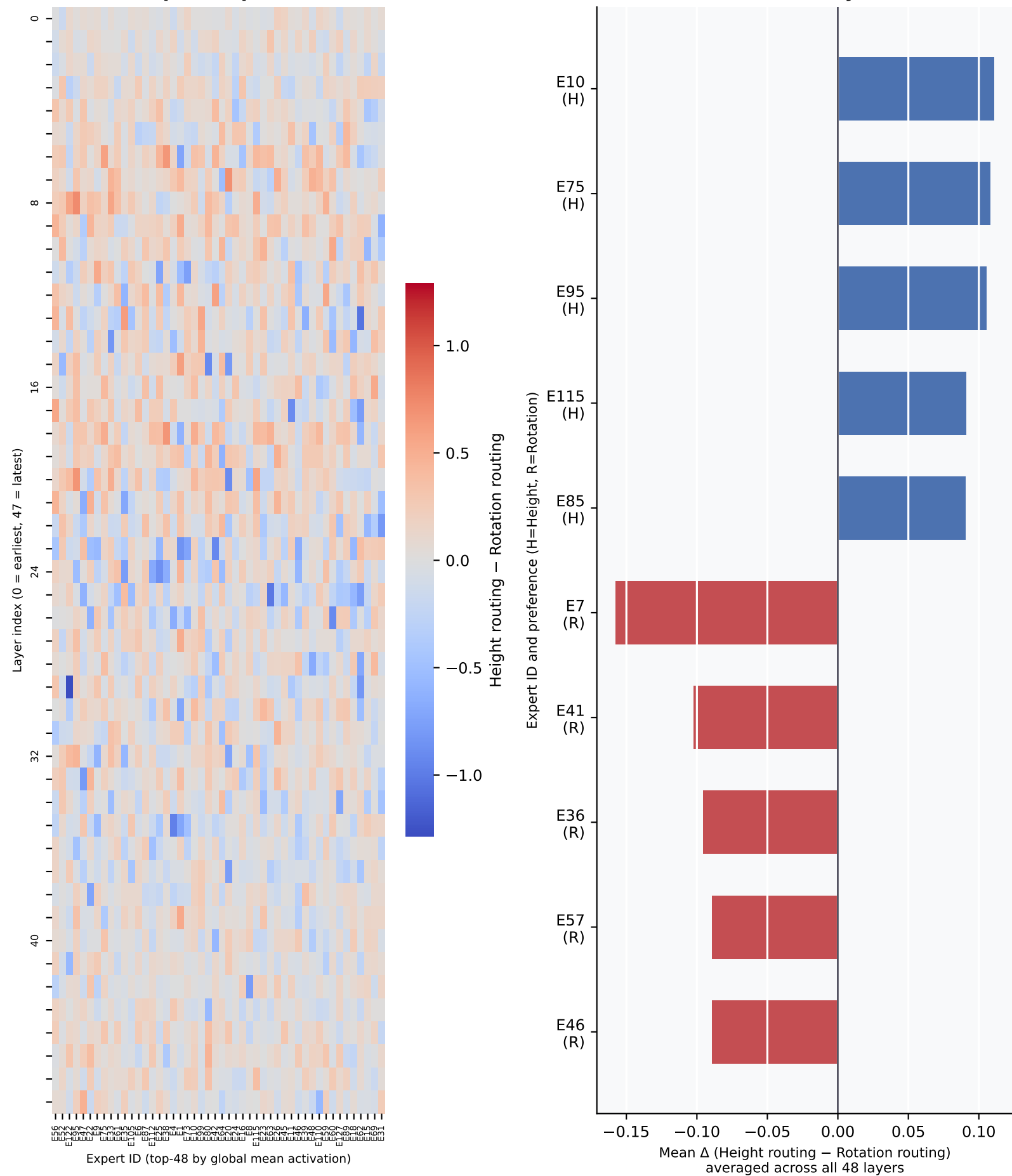


Height vs Rotation — Expert Specialisation

Blue = Height-preferring | Red = Rotation-preferring | White = shared

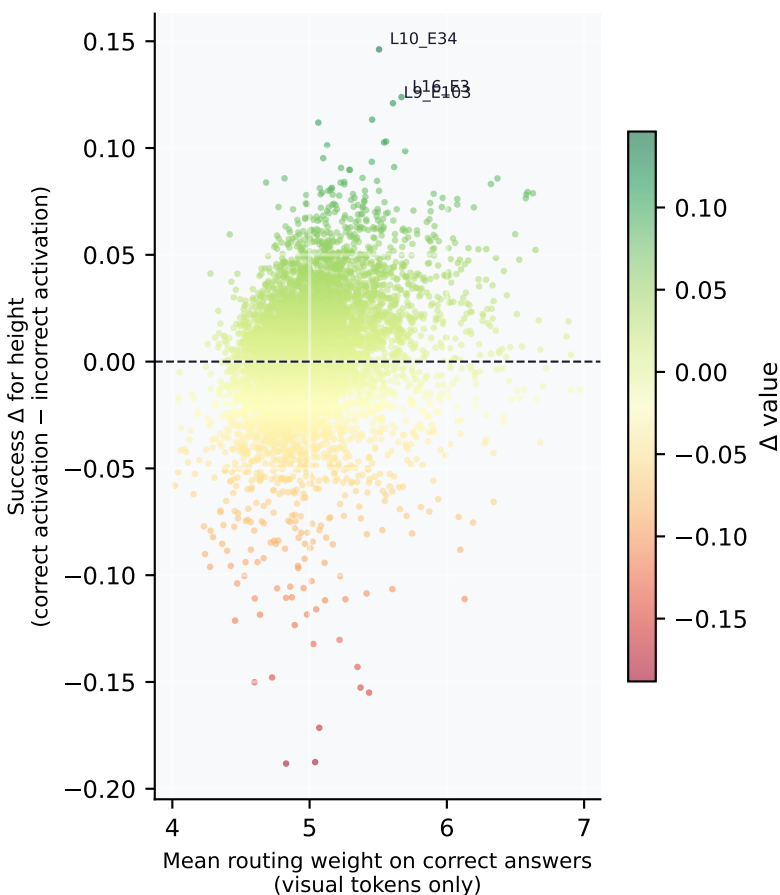
Height – Rotation difference (top-48 experts)

Most specialised experts across all layers

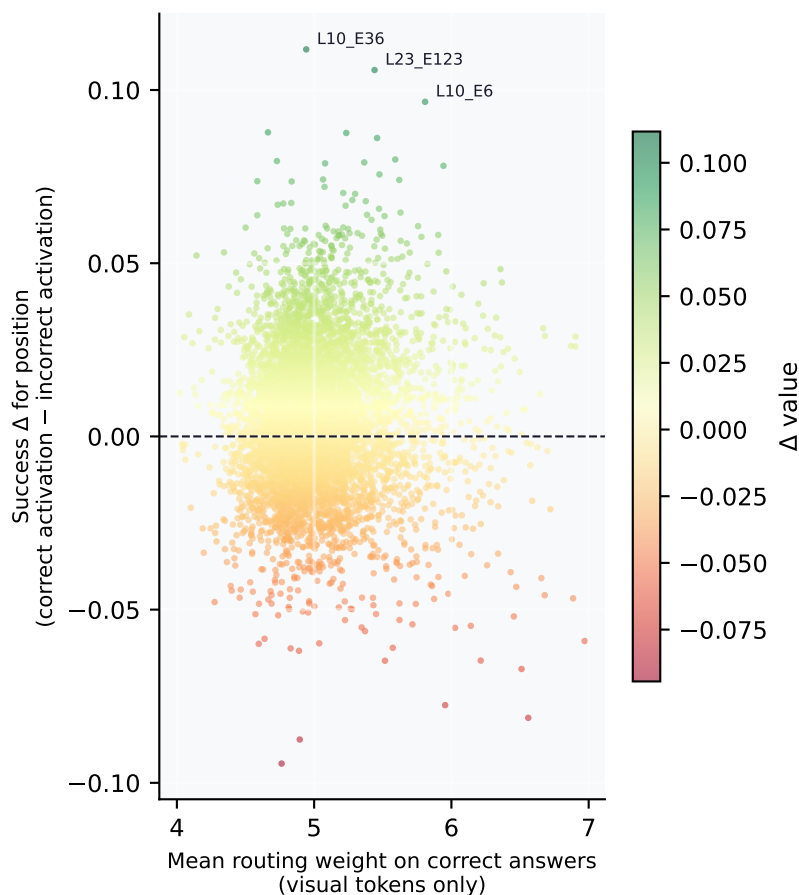


Expert Activation vs Success Δ — per Category (Run 2)
Top-right quadrant = high activation + positive correlation with correct answers

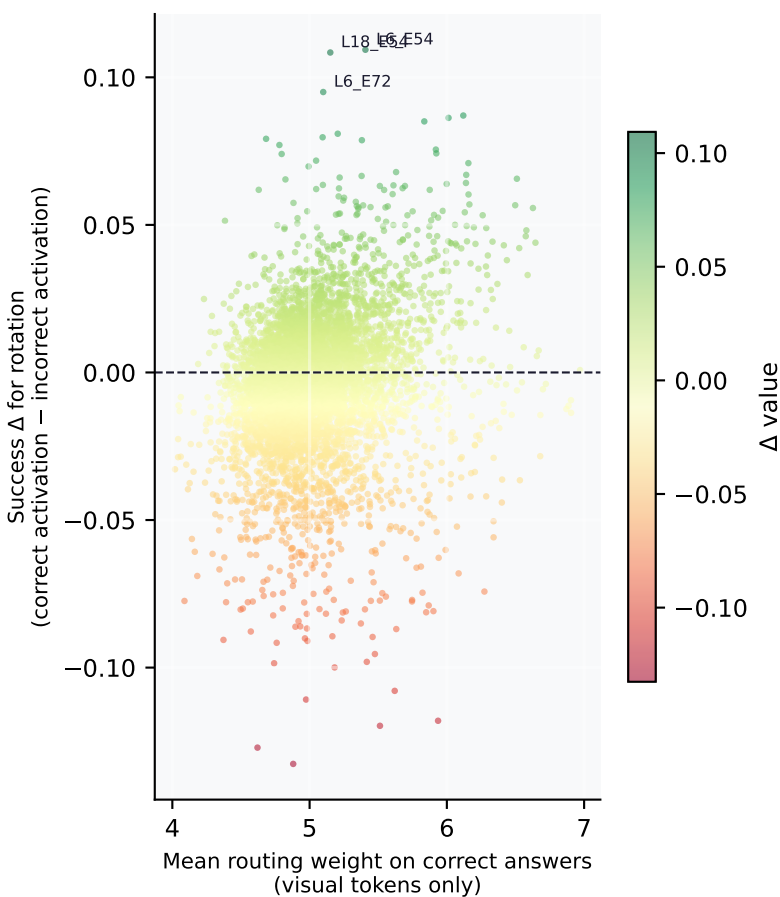
HEIGHT



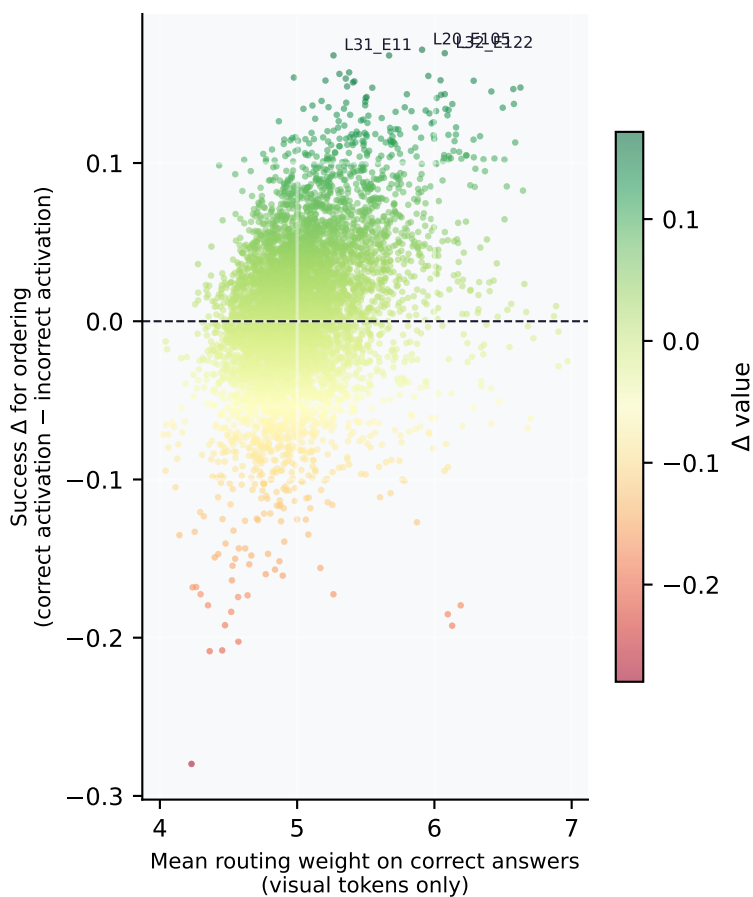
POSITION



ROTATION



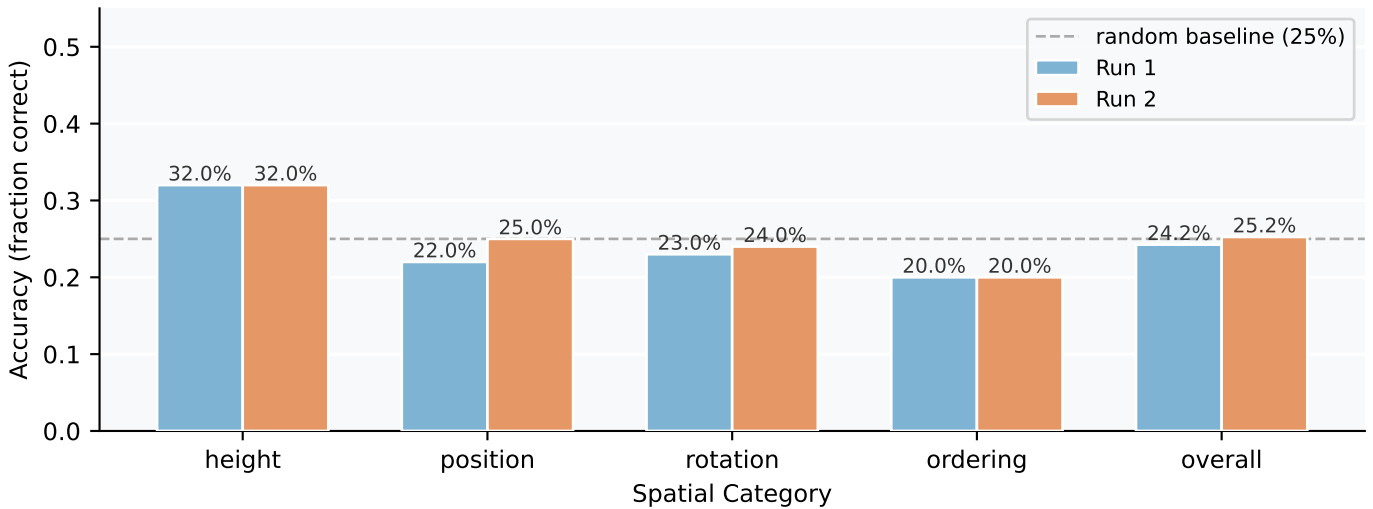
ORDERING



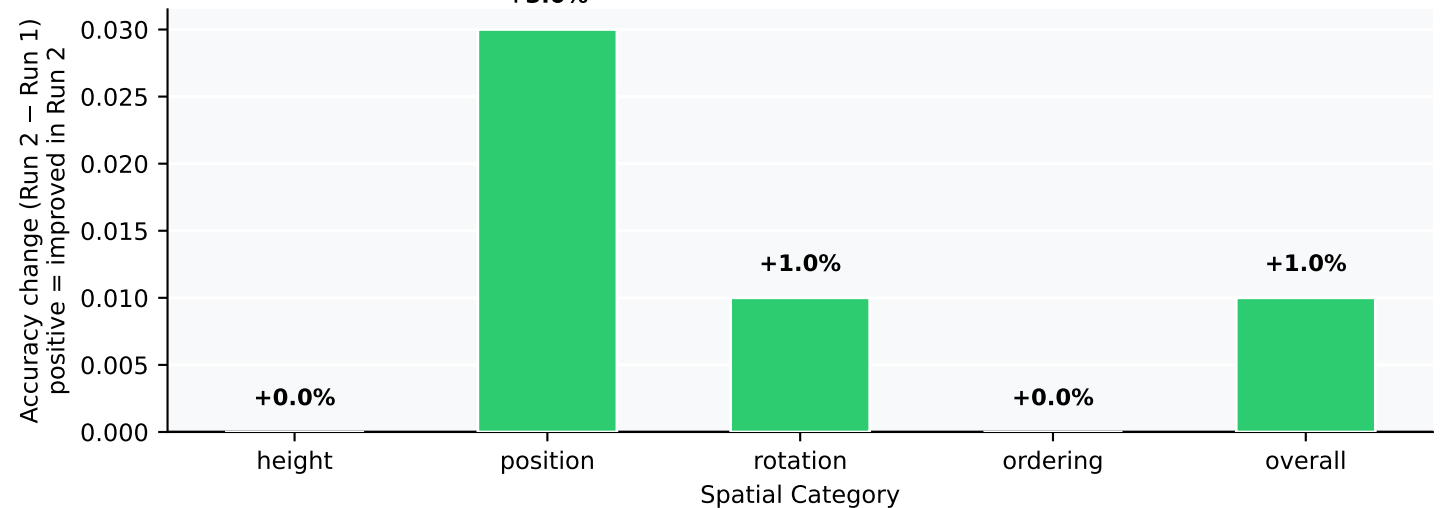
Run 1 vs Run 2 — Repeatability Analysis

Same model, same dataset, same prompts — measuring variance across runs

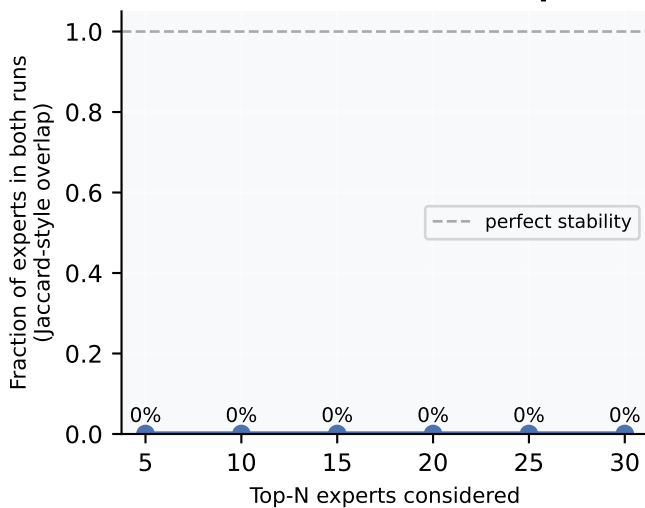
Per-Category Accuracy: Run 1 vs Run 2



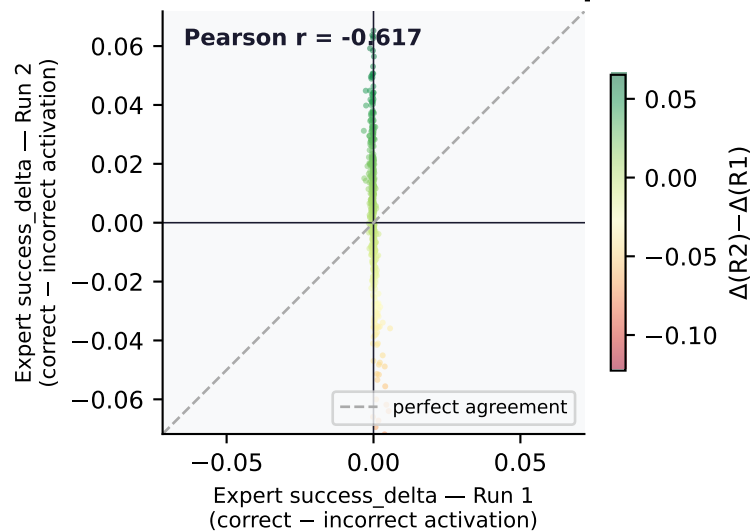
Accuracy Delta: Run 2 – Run 1



Expert Leaderboard Stability Run1 n Run2 overlap



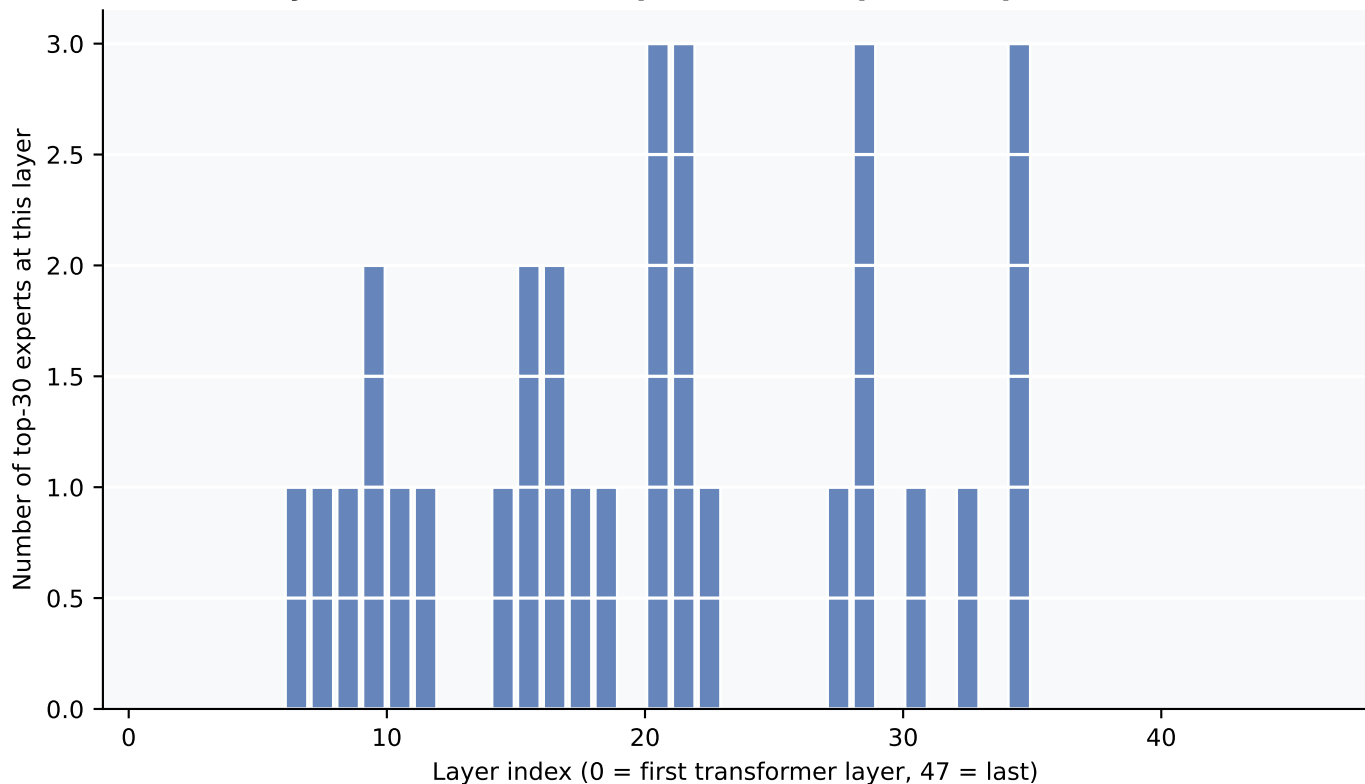
Expert Δ Correlation Run 1 vs Run 2 (500 random experts)



Where in the Network Do Spatial Experts Live?

Layer distribution of the top-30 global spatial experts vs category specialists

Layer Distribution — Top-30 Global Spatial Experts (Run 2)



Layer Distribution — Per-Category Top-10 Specialist Experts (Run 2)

